

Testing transfer effects of domain-general cognitive abilities to domain-specific decision making in Soccer: Are basic cognitive abilities related to creativity and intelligent decision making?

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ABSTRACT

A frequently investigated question is how domain-general cognitive abilities are related to domain-specific performance. The present study addresses several shortcomings of earlier research on the hypothesized positive relationship between domain-general cognitive ability tests and domain-specific cognitive skills (i.e., decision making) in soccer. By measuring both domain-general cognitive abilities (attentional breadth, multiple object tracking, working memory, and inhibition) and domain-specific performance (creative and intelligent decision making) on continuous scales ($n > 100$ participants [competitive soccer players] for every pairwise correlation), we avoid the common pitfalls of cross-sectional, between-group comparisons in this field. The results reveal only a small-to-medium correlation between multiple object tracking and intelligent decision making in soccer and a small correlation between attentional breadth and creative decision making, with no other significant associations found in the remaining six hypothesis tests. In addition, playing level was not associated with domain-general cognitive abilities. Taken together, these findings contribute to the growing skepticism regarding the role of domain-general cognitive abilities in explaining team-sport expertise.

Successful performance in fast-paced team sports such as soccer depends on a broad range of physical, technical, and cognitive abilities. While physical and technical skills have traditionally received most attention, interest in cognitive abilities (e.g., executive functions, attention; Vona et al., 2024) and cognitive skills (e.g., decision making and anticipation; Kalén et al., 2021) has increased substantially. A recent debate in the Proceedings of the National Academy of Sciences (Bonetti et al., 2025a, Bonetti et al., 2025b; Musculus et al., 2025) highlights a central issue in this literature: whether domain-general cognitive abilities contribute to domain-specific soccer performance. Accordingly, the present study addresses methodological shortcomings identified in prior research (Musculus et al., 2025) by examining the relationship between domain-general cognitive abilities and domain-specific soccer performance using continuous measures of attentional breadth, multiple object tracking, working memory, inhibition, and soccer decision making.

More broadly, this question concerns how, domain-general cognitive abilities relate to specialized, domain-specific skills (Fransen, 2024; Furley et al., 2023). The issue is rooted in expertise research (Ericsson et al., 1993, 2006) and historically linked to theories of human

intelligence (Binet & Simon, 1905; Galton, 1883; Terman, 1916). Although definitions vary (Furley et al., 2023; Kalén et al., 2021; Vona et al., 2024), domain-general abilities are typically conceptualized as broad cognitive processes applicable across contexts, whereas domain-specific skills are specialized competencies acquired through extensive deliberate practice and largely confined to the domain in which they develop (Ericsson et al., 2006).

The appeal of domain-general cognitive testing stems from evidence linking such measures to important outcomes, including academic and occupational performance (e.g., Schmidt & Hunter, 1998). Inspired by findings connecting executive functions and intelligence to desirable life outcomes (Deary et al., 2007; Diamond, 2013), sport researchers have increasingly investigated whether basic cognitive abilities are associated with athletic performance (Bonetti et al., 2025a, Bonetti et al., 2025b; Furley et al., 2023; Kalén et al., 2021; Knöllner et al., 2022; Memmert, 2009; Musculus et al., 2025; Vona et al., 2024). The underlying assumption is that elite sport performance requires efficient attentional control, inhibition, and concentration under demanding conditions; therefore, successful athletes should outperform less successful athletes or non-athletes on domain-general cognitive tests.

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Evidence from several meta-analyses appears consistent with this assumption, showing associations between sport expertise and enhanced performance on domain-general cognitive tasks (Kalén et al., 2021; Logan et al., 2023; Scharfen and Memmert, 2021; Voss et al., 2010)—although some of these meta-analyses conclude that domain-specific cognitive skills are more important in explaining sport expertise than domain-general cognitive measures. Moreover, some studies suggest that laboratory-based cognitive measures predict sport-specific skills and performance (e.g., Cona et al., 2015; Vestberg et al., 2017). This perspective, termed the cognitive component skill approach (Voss et al., 2010), challenges influential expertise theories (e.g., Eccles, 2006; Ericsson et al., 2006), which posit that expertise is largely domain-specific and does not transfer broadly across contexts. Despite some support for the cognitive component skill approach, findings remain inconsistent. Several studies report no relationship between domain-general cognitive abilities and domain-specific sport skills, and reviews have highlighted substantial methodological weaknesses in the primary literature (Furley et al., 2023; Kalén et al., 2021). The present research therefore addresses some of these limitations that mainly arise through the dominant use of cross-sectional expert–novice comparisons in this field of research.

1. Primary research on the relationship between domain-general measures of cognitive abilities and domain-specific performance in team-sports

Research in sport science has long examined whether domain-general cognitive abilities—particularly executive functions—relate to sport performance. A highly cited study by Vestberg et al. (2012) reported that professional soccer players outperformed lower-level players and a normative population on the Delis-Kaplan Executive Function System (D-KEFS, Delis et al., 2001). Executive-function scores also predicted future on-field performance (goals and assists two years later). Based on these findings, the authors argued that team-sport expertise may translate to general cognitive domains and reflect strengths in spatial attention, divided attention, working memory, and mentalizing capacity. Subsequent studies have similarly reported positive associations between domain-general cognitive measures and elite team-sport performance (e.g., Elferink-Gemser et al., 2018; Huijgen et al., 2015; Romeas et al., 2016; Verburgh et al., 2014; Vestberg et al., 2017, 2020).

However, overall empirical support for this relationship remains weak. Many studies rely on small samples, cross-sectional designs, inadequate correction for multiple comparisons, and selective reporting of significant findings from larger cognitive test batteries. Correspondingly, several studies have failed to replicate these associations and concluded that domain-general cognitive tests are not reliably linked to superior athletic performance (Beavan et al., 2020a, 2020b; Furley & Memmert, 2010; Memmert et al., 2009; Vona et al., 2024). Not surprisingly there remains continued debate about the role of domain-general cognitive abilities in contributing to domain-specific sport expertise (Bonetti et al., 2026; Fitzgerald et al., 2026; Heisler et al., 2023; Knöbel et al., 2025; Seidel-Marzi et al., 2026).

Given these inconsistencies, the present research aimed to address key methodological limitations in testing the hypothesized association between domain-general cognitive abilities and domain-specific sport performance arising from the dominant use of cross-sectional expert–novice comparisons in soccer and related sports (e.g., Kalén et al., 2021; Musculus et al., 2025; Scharfen & Memmert, 2019; Voss et al., 2010).

2. Pitfalls of previous research

Confounding variables. From a classical test theory perspective, cross-sectional between-group designs estimate the ratio of systematic to unsystematic variance. In this context, systematic variance reflects shared variance between cognitive performance and athletic expertise,

whereas unsystematic variance reflects uncontrolled influences. Because participants are not randomly assigned to expertise groups, observed differences cannot establish causality and may instead reflect confounding variables associated with elite sport participation.

Elite performance is influenced by numerous interacting factors beyond cognition (Hambrick et al., 2016). For example, aerobic fitness is positively associated with cognitive functioning (Hillman et al., 2008; Kramer et al., 1999), raising the possibility that cognitive advantages in elite athletes reflect superior fitness rather than expertise itself. Yet many studies omit high-fitness control groups (e.g., Chaddock et al., 2011; Vestberg et al., 2012). Additional confounds may include ambition, grit, and stress resilience. As noted by Adele Diamond (2020, p. 221), cognitive tests are imperfect indicators of underlying ability, and low or high scores may arise from factors unrelated to the targeted construct.

Defining expertise. A further limitation concerns the inconsistent definition of expertise (Ericsson, 1996; Posner, 1988), which gives researchers substantial flexibility in defining “expert” and “novice” groups. Swann et al. (2015) highlighted this inconsistency and proposed that expertise should be conceptualized along a continuum rather than as a simple expert–novice dichotomy. To improve methodological rigor, the authors introduced a taxonomy for classifying athlete expertise levels based on factors such as highest performance standard, success, experience, and competitiveness.

Choice of cognitive measures. The suitability of domain-general cognitive tests for sport research has also been questioned. Cremen and Carson (2017) and Furley et al. (2023) argued that many commonly used measures, including the D-KEFS (Delis et al., 2001), were developed for clinical rather than athletic populations and may show limited psychometric validity in performance settings (Baron, 2004; Schmidt, 2003). Nyongesa et al. (2019) similarly identified reliability and validity concerns, particularly in adolescent samples. Moreover, Finkenzeller et al. (2021) reported poor test–retest reliability for the design fluency test used in Vestberg et al. (2012, 2017, 2020), concluding that it is unsuitable for sport applications. Collectively, these findings suggest that standard domain-general assessments may lack sufficient psychometric rigor for sport-performance research.

Small sample sizes. Finally, elite-athlete research is frequently constrained by very small sample sizes, increasing the risk of Type I and Type II errors (Schweizer & Furley, 2016). For example, Vestberg et al. (2012) included groups of only 11–17 participants. Assuming a medium effect size (Cohen's $d = .5$), the statistical power for comparisons between elite and lower-level players was approximately .27 for men and .22 for women; even after combining groups, power increased only to .46. Thus, the study had less than a 50% probability of detecting a true medium-sized effect. As noted by Fraley and Vazire (2014), such low-powered studies approach chance-level sensitivity.

3. The present research

Given the problems with primary studies that seemingly support the cognitive component skill approach we took a step back and addressed a more fundamental question: Are domain-general cognitive ability tests that should theoretically be relevant for successful soccer decision-making actually related to measures of game creativity and game intelligence in a soccer-specific decision-making test? In line with recommendations from the perceptual-cognitive skill framework by Hadlow et al. (2018), we measured both domain-general cognitive abilities and domain-specific performance on continuous scales. This approach allowed us to avoid many of the pitfalls associated with typical cross-sectional between-group comparisons (e.g., Bonetti et al., 2025a). Furthermore, we ensured that the domain-general cognitive ability tests and the soccer decision-making tests were theoretically relevant to intelligent and creative decision-making. Accordingly, the present study does not claim to fully resolve the broader methodological challenges historically affecting research on cognitive abilities and sports expertise;

rather, it aims to incrementally address specific limitations of expert-novice comparisons while remaining transparent about the constraints inherent to its cross-sectional and correlational approach.

We acknowledge that the logical argument for overlap between domain-general cognitive ability measures and the cognitive demands of soccer appears plausible. Specifically, we considered it feasible that reliable measures of attentional breadth and multiple object tracking would show overlap with creative and intelligent soccer decision-making as this should allow to integrate more relevant objects (teammates and opponents) into the decision-making process. We also hypothesized that measures indicating an individual's ability to inhibit task-irrelevant information and focus attention in a goal-directed manner should correlate with decision-making in soccer—by e.g., inhibiting “irrelevant” information and “decision options”. Thus, we carefully selected domain-general cognitive ability tests and soccer-specific decision-making measures based on the quote by Vestberg et al. (2012, p. 1): “a good team player could be characterized by excellent spatial attention, divided attention, and working memory.” To that end, we selected a battery of four computer-based cognitive ability tests designed to measure the full range of cognitive abilities described by Vestberg and colleagues and previously used in soccer-related cognitive testing (Memmert, 2021).

To evaluate the capacity for broad visual attention and complex information integration, two domain-general tasks were employed: the Attentional Window (AW) task (Hüttermann and Memmert, 2015; Hüttermann et al., 2014) and Multiple Object Tracking (MOT; Alvarez & Franconeri, 2005). The AW task measures the maximum spatial distribution of visual focus—a metric previously shown to differentiate expert soccer players from non-athletes (Hüttermann et al., 2019). Similarly, MOT assesses the simultaneous tracking of multiple dynamic stimuli, a core attentional component with high face validity in sport (Faubert, 2013; Memmert et al., 2009). Although empirical evidence regarding the MOT-expertise link remains equivocal (Vater et al., 2021), both tasks conceptually overlap with the perceptual-cognitive demands of soccer. Consequently, we hypothesized that superior performance in both AW and MOT would positively correlate with intelligent and creative decision-making. Working Memory Capacity (WMC) was assessed as a proxy for executive attentional control (Conway et al., 2005). WMC is strongly associated with high-level cognitive processes, including language comprehension and reasoning, largely due to the shared requirement for endogenous attentional regulation (King & Just, 1991; Kyllonen & Christal, 1990). Given that effective decision-making in soccer necessitates the simultaneous maintenance and processing of dynamic environmental information, we hypothesized a positive correlation between WMC and decision quality (cf. Vestberg et al., 2012). This hypothesis was maintained despite ambiguous empirical findings with some studies showing no significant association between WMC and tactical decision-making in team sports (Furley & Memmert, 2012, 2015).

A response competition paradigm (Beck & Lavie, 2005), adapted for soccer contexts (Furley et al., 2013), was employed to assess domain-general cognitive inhibition. This task evaluates an athlete's capacity to maintain focus on task-relevant stimuli while suppressing task-irrelevant distractors under varying levels of perceptual load. While previous research has not established a definitive link between performance on this specific paradigm and soccer expertise (Furley et al., 2013), its inclusion is theoretically justified: both intelligent and creative decision-making in soccer necessitate the efficient integration of pertinent cues and the simultaneous inhibition of environmental noise.

For domain-specific decision-making, we adapted the divergent and convergent decision-making tasks of Memmert (2011) for the SoccerBot360—a circular, immersive video display platform developed by Umbrella Software (see Fig. 1). This setup, with high trial counts (Schweizer et al., 2020), allows for a reliable assessment of both creative and intelligent soccer decision-making in a soccer-specific environment. Although this setup is advertised as a representative soccer training and

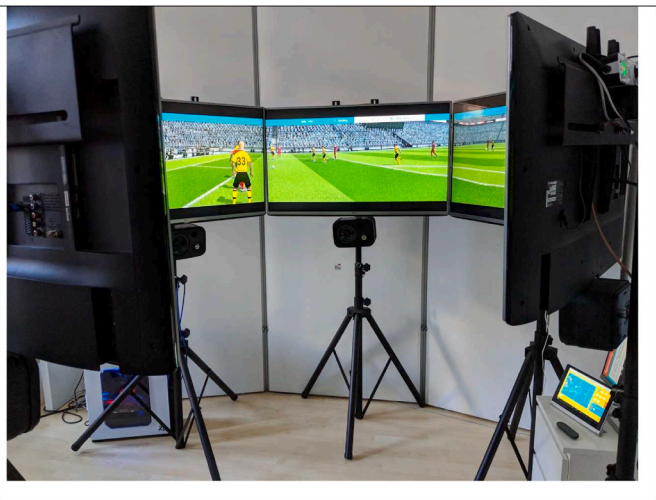


Fig. 1. Illustration of the domain-specific soccer decision making task and sample stimulus material used in the present research.

assessment system (<https://www.soccerbot360.de/>) and has been used in several research papers (Heilmann et al., 2021; Knöbel et al., 2025; Musculus et al., 2022), we initially attempted to validate our assumption that our soccerbot360 decision making task taps into domain-specific, perceptual-cognitive expertise. In this endeavor, we compared the quality of decisions of two groups of soccer players against top-rated decision options of a panel of expert A-license soccer coaches. Supporting our assumption, lower-level soccer players ($n = 320$, everything below the 5th highest soccer league in Germany) showed significantly worse decisions ($p < .001$) than higher level soccer players ($n = 159$; everything from the 5th highest league; 5th to 1st league). There also was a significant Spearman correlation between playing level and the quality of decisions across all 470 soccer players ($p = .018$).

4. Method

4.1. Participants

A total of 1053 participants ($M_{age} = 23.97$, $SD\ 3.74$; 93.3% male, 6.7% female); took part in this study. All participants were competitive soccer players with 8 playing in the 1. German league, 1 in the 2nd, 3 in the 3rd, 21 in the fourth, 134 in the fifth, 191 in the 6th, 18 in the 7th, 215 in the 8th, 317 in the 9th, and 11 in the 10th German League (134 players did not disclose this information). Only outfield players were included in the sample that played in positions similar to the decision-making scenario in this research (i.e., no goal-keepers or strikers). Table 1 gives an overview of how participants were assigned to domain-general and domain-specific cognitive tests. A subsample of 101 participants is excluded from Table 1 because they did not complete a

Table 1
Sample Size for all tested combinations of cognitive ability tasks and Soccerbot tasks.

Soccerbot_Task	Cognitive Ability_Task	n	Detectable correlations with a power of .8 and alpha .05 (.005)
Game Creativity	Attention Window	121	.22 (.30)
Game Creativity	Multiple Object Tracking	130	.21 (.29)
Game Creativity	Inhibition	100	.24 (.33)
Game Creativity	Working Memory	140	.21 (.28)
Game Intelligence	Attention Window	120	.22 (.30)
Game Intelligence	Multiple Object Tracking	101	.24 (.33)
Game Intelligence	Inhibition	120	.22 (.30)
Game Intelligence	Working Memory	120	.22 (.30)

domain-general cognitive test; instead, they performed both the game intelligence and game creativity tasks to examine the bivariate relationship between these two domain-specific measures. Bivariate correlation analysis revealed a significant positive association ($r = .301$, $p = .002$), indicating that while the two measures are moderately related, they capture distinct underlying constructs. To accommodate participant time constraints—which limited each testing session to a maximum duration of 1.5 h—a counterbalanced, permuted-block randomization procedure was implemented. Each participant was assigned to complete one of two domain-specific soccer decision-making tasks (game intelligence or creativity) paired with one of four domain-general cognitive tasks (Attention Window, Multiple Object Tracking, Inhibition, or Working Memory), yielding eight distinct test pairings.

To verify the efficacy of the randomization procedure, preliminary analyses were conducted to compare the eight task-pair subsamples on key demographic and athletic characteristics. Results confirmed that the groups did not significantly differ on any baseline variables (e.g., competitive playing level) or the soccer-decision-making variables (all $p > .05$), indicating that the random allocation successfully produced comparable subsamples and that the observed correlational patterns are not attributable to systematic sampling bias across task pairings. Participants were only included in the analyses if data was complete for one domain-specific decision-making test and one domain-general cognitive ability test. Participants had normal or corrected-to-normal vision and provided informed consent before participating. The study was approved by the relevant ethics committee (Nr. 14/2024).

A further remedy of the present research was to have at least decently powered statistical hypotheses tests as this has been identified as a big problem when testing for the role of domain-general cognitive ability tests in team-sports expertise (Furley et al., 2016, 2023; Schweizer and Furley, 2016). Therefore, we sampled over 1000 participants to have at least an n of 100 for every pairwise correlation between domain-general cognitive ability test and domain-specific decision-making test. This sampling procedure ensured that every one-tailed Pearson correlation test had a sensitivity to reliably detect small-to-medium correlations ($r = .24$) with a power of .8 and a significance level of .05. See Table 1 for the sensitivity analyses for every individual test. However, given concerns relating to the inflation of type one error risk with multiple hypotheses testing, the Holm-Bonferroni technique was used to adjust the p -values. Therefore, the adjusted tests could only reliably detect larger correlations (see brackets in the last column of Table 1) and we additionally run Bayesian correlations as these do not rely on the same power assumptions.

4.2. Procedure

Participants completed one out of four computer-based cognitive ability measures, and one out of two game-based performance assessments inside the SoccerBot360 system—i.e., every participant completed one domain-general cognitive ability test and one domain-specific soccer decision-making test. The presentation order of the domain-general cognitive test and the domain-specific soccer decision-making task was counterbalanced across participants to control for potential order or fatigue effects. The study was conducted in a controlled laboratory setting, and all tasks were administered within a single session.

4.3. Cognitive ability measures

Attention Window Task (AWT): The Attention Window Task (AWT) was used to assess participants' ability to distribute attention across peripheral areas while being instructed to maintain central fixation, an important cognitive skill in dynamic environments such as team sports (Hüttermann et al., 2014). On each trial, two stimuli appeared for 12 ms along a randomly selected meridian (horizontal, vertical, diagonal), symmetrically positioned around a central fixation cross, at

separations ranging from 2° to 60° visual angle. Each stimulus consisted of four elements varying in shape and color, with 0–4 light gray triangles defining the targets. After stimulus presentation, a backward mask was displayed for 100 ms to reduce afterimages and limit visual persistence. Participants were instructed to fixate centrally and report the number of triangles in both peripheral stimuli. The task probed participants' capacity to spread attention over increasingly large areas of the visual field.

The dependent variable was **attentional breadth (θ)**:

$$\theta = \sqrt{\frac{1}{N} \sum_{i=1}^N E_i^2}$$

where E_i is the eccentricity (in degrees) corresponding to 75% correct detection on meridian i , averaged across meridians ($N = 3$). This threshold defines the size of an individual's "attention window." Previous work demonstrated that this operationalization yields reliable individual differences (split-half reliability $\approx .77-.82$; Hüttermann et al., 2014).

Additional methodological detail can be found in Hüttermann et al. (2014).

Multiple Object Tracking (MOT). The MOT paradigm is a classic measure of dynamic visual attention capacity (Alvarez & Franconeri, 2007). In each trial, 16 identical green circles appeared on a black background, and a subset of 4 targets were briefly cued by flashing red for 2 s. After cueing, all objects became identical, and participants tracked the target objects for 5 s as they moved randomly across the display, bouncing off each other and display edges while maintaining minimum separation (4° center-to-center). Participants were instructed to fixate centrally throughout. At motion offset, participants identified which items were the original targets.

To quantify tracking performance, an adaptive **1-up/1-down staircase procedure** was used to titrate object speed. After correct responses, speed increased by .5°/s; after errors, it decreased by .5°/s. The staircase converged on the participant's **threshold tracking speed (S)**:

$$S = \text{mean speed at 75\% accuracy}$$

Thresholds reflect the highest speed at which an individual could track 4 targets reliably.

The paradigm has strong psychometric properties, with test–retest reliability around $r = .80$ (Alvarez & Franconeri, 2007). Detailed task parameters are provided in Alvarez and Franconeri (2007).

Working Memory Capacity (WMC). Working memory was assessed using a computerized counting span task, which simultaneously requires attention control and storage, making it a well-validated measure of domain-general executive function (Conway et al., 2005). Participants viewed displays containing intermixed targets (dark blue circles) and distractors (green circles, dark blue squares). In each display, they counted the number of target items aloud and reported the total immediately after viewing. A series of 2–7 such displays formed a set, after which participants recalled the sequence of counts in order.

Performance was scored using the partial credit unit (PCU) scoring method recommended by Conway et al. (2005), which gives proportionate credit for partial recalls:

$$PCU = \frac{1}{K} \sum_{j=1}^K \sum_{\{i=1\}^c} \{n_j\}^c \left\{ \sum_{j=1}^K n_j \right\}$$

Where C_{ij} if count i in set j was recalled correctly in its serial position, 0 otherwise; n_j is the set size; K is the number of sets. This yields a continuous score reflecting overall working memory capacity as a percentage of correct recalls.

This task is widely used due to its high reliability (Cronbach's $\alpha \approx 0.78-.85$) and robust construct validity as a measure of executive attention capacity (Conway et al., 2005). Further procedural details can be found in Conway et al. (2005).

Perceptual Load (PL) Task. The Inhibition was measured using a soccer-adapted perceptual load task (Furley et al., 2013), building on Lavie's (1995) load theory of selective attention. Participants viewed static soccer scenes (with players of a team being portrayed schematically as Xs or Os) following a 1000 ms fixation cross. Displays were presented for 100 ms and varied in complexity (low-load: fewer players; high-load: many players). The primary task was to identify which player possessed the ball as quickly and accurately as possible. A task-irrelevant distractor appeared at fixation concurrently with the soccer scene. The distractor was either the same as the player in possession of the ball—an O if a O-Player was in possession of the ball—or different—an X if an O-player was in possession of the ball. On response-congruent trials, the distractor matched the target response; on incongruent trials, it differed.

Perceptual load was manipulated across blocks, allowing assessment of distractor interference under varying attentional demands.

The **dependent variable was the Inhibition Score (IS)**, defined as the difference in mean reaction time between incongruent i and congruent c trials under low-load conditions and high-load conditions:

$$IS = \overline{RT}_{\{i,LL\}} - \overline{RT}_{\{c,LL\}}$$

where *RT* is the reaction time for low-load trials *LL* in the *i* (incongruent) and *c* (congruent) conditions. The formula would be equivalent for high-load conditions with the replacement of *HL* for *LL*. In alignment with cognitive load theory, which posits that distractor vulnerability is enhanced under low-load conditions (Furley et al., 2016; Lavie, 1995), our analysis focused solely on the low-load index of inhibition efficiency. This approach also served to constrain the number of statistical comparisons within the current paradigm. Crucially, this metric provides a clean isolation of inhibition efficiency. The task has shown moderate reliability in sport contexts ($ICC \approx .70$ for RT_{LL} ; Furley et al., 2013). A full description of the paradigm is available in Furley et al. (2013). Crucially, the inhibition score is operationalized as the reaction time cost ($RT_{incongruent} - RT_{congruent}$), such that lower scores reflect smaller distractor interference and superior inhibition efficiency, whereas higher scores reflect greater interference and poorer inhibition efficiency. Consequently, a negative correlation between this interference score and domain-specific decision-making performance indicates a positive functional relationship—namely, that superior inhibitory control is associated with better decision-making.

4.4. Game intelligence and creativity measures

To assess soccer-specific cognitive skills, participants completed two types of decision-making tasks using the Soccerbot360 system (<https://www.soccerbot360.de/>). This innovative video display platform includes seven high-definition screens arranged in a circular format around the participant, simulating a 360° game environment. The system enables immersion in realistic soccer scenarios with controlled, repeatable testing conditions. The rationale for choosing the measures described below was that these paradigms have been shown to have good psychometric properties for measuring both divergent/creative (Johnson & Raab, 2003; Memmert, 2011) and convergent/intelligent thinking (Memmert, 2010) in team sports like soccer.

Game intelligence. For the game intelligence condition, participants stood at the center of the Soccerbot360 setup and were shown 30 authentic Bundesliga match scenes. Each scene was paused at a critical decision moment and the last frozen frame stayed visible for the remaining decision time (cf. Memmert, 2010). Although, this approach arguably reduces the representativeness of the decision-making task by reducing the time constraints of the decision-making scenario, it has been standard praxis when assessing convergent (game intelligence) and divergent (creative) thinking in sports using option generation paradigms (cf., Hüttermann et al., 2019; Johnson & Raab, 2003; Memmert, 2010, 2011). Every freeze frame was a scene of a game opening situation

in the midfield. Participants were then instructed to select the most tactically appropriate passing option within 30 s (shot and dribbling were no options in the game intelligence scenes). Their chosen decision option was communicated verbally (potentially assisted by pointing to a respective player) for passing decisions and recorded by the experimenter. All potential decision options for every respective scene had been independently scored on a scale of 1 to 7 by two expert A-license soccer coaches. Only scenes were selected for which the two soccer experts had given the highest score for the same passing option. For example, expert A gave a 6 for a pass to player X and expert B 5 points (both giving their highest score for pass to player X), then a pass to player X had a value of 5.5 (average between the two experts; agreement between experts (i.e., inter-observer reliability) showed high levels of agreement for all passing options of all 30 scenes ($r_s = .73$ to $.94$, $p_s < .001$). The scoring system was based on the tactical quality of the decision, with higher scores representing more strategically intelligent actions. The final game intelligence score was calculated as the average across all scenes.

Game creativity. In the creativity condition, participants were again presented with 24 different game scenes (last frame visible for 45 s; (cf., Hüttermann et al., 2019; Johnson & Raab, 2003; Memmert, 2010, 2011) but were asked to name all potential decision options (this time including dribbling and shot option besides passing options to certain players) in an offense position that they believed could plausibly result in a goal-scoring opportunity. This option generation paradigm has been the standard paradigm in assessing tactical creativity in sport because it directly operationalizes divergent thinking within a sporting context. While expanding the response criteria to include multiple action modalities (e.g., passing, dribbling, shooting) and extending the time window to 45 s removes the typical time constraints of real-world soccer decision-making, it crucially enables researchers to capture the full spectrum of a participant's solution space. Specifically, this approach measures fluency (quantity of options), flexibility (number of different categories in the chosen options) and originality (uniqueness of options)—the core behavioral markers of tactical creativity in dynamic team sports. Because tactical responses within a given game scene recur frequently across participants, all suggested actions were first cataloged into a set of unique, distinct decision options per scene. Each unique decision option was then independently evaluated by the same two expert A-license soccer coaches (who had received prior training in this option-generation paradigm and a rubric-tool with 5 categories [pass, chip, dribbling, shot, other] for the evaluation of flexibility) based on novelty, tactical soundness, and effectiveness. Individual participant responses were subsequently mapped onto these evaluated options to derive individual metrics. The overall mean Originality score was 3.53 ($SD = .18$), Flexibility = 2.35 ($SD = .55$), and Fluency = 3.34 ($SD = .97$). Inter-observer reliability between the two experts showed decent levels of agreement across the unique decision options of all 24 scenes ($r_s = .53$ to $.97$, $p_s < .001$). Creativity metrics were computed identically as in Furley and Memmert (2015, see for full details). Specifically, to derive the final outcome variable entered into the correlation analyses, a composite creativity score was calculated by averaging the z-transformed values of fluency, flexibility, and originality. All testing took place under standardized laboratory conditions.

4.5. Data analysis

To examine the relationship between cognitive abilities and soccer-specific decision-making, Pearson product-moment correlations were computed between each domain-general cognitive ability measure and the Soccerbot game intelligence and creativity score. In addition, we calculated Spearman correlations between playing level and the cognitive ability measures. All Pearson correlations were reported with their respective 95% confidence intervals. Given concerns relating to the inflation of type one error risk with multiple hypotheses testing, the Holm-Bonferroni technique was used to adjust the p-values from the 8

bivariate correlations for our main hypothesis tests regarding the association between domain-general cognitive ability measures and domain-specific decision-making skills reported above (Holm, 1979). To quantify the strength of evidence for the presence or absence of a correlation between domain-specific decision making and domain-general cognitive abilities, we additionally conducted Bayesian Pearson correlations in JASP. Evidence was evaluated using the Bayes Factor (BF_{01}), which represents the ratio of the likelihood of the data under the null hypothesis (H_0) relative to the alternative hypothesis (H_1). Following established conventions (Lee & Wagenmakers, 2014) BF_{01} values were interpreted as follows: evidence for H_0 : moderate (3–10) and strong (>10); evidence for H_1 : moderate (.10–.33) and strong ($<.10$). All data were analyzed with R, JASP, and SPSS.

5. Results

Table 2 summarizes the main hypothesis tests regarding the correlations between the domain-general cognitive ability measures and game intelligence and game creativity.

5.1. Game intelligence analyses

As can be seen in Table 2, there was a significant correlation between Multiple Object Tracking and Game Intelligence ($r = .38, p = .008$). This small-to-medium correlation is indicative that soccer players that were better at tracking multiple objects at higher speeds also showed slightly better game intelligence. The Bayes analyses back up this inference by showing strong support for the alternative hypothesis that tracking multiple objects is correlated with game intelligence. Fig. 2 visually summarize the main hypotheses tests regarding the correlations between the domain-general cognitive ability measures and game intelligence.

5.2. Game creativity analyses

Regarding game creativity results showed a small to moderate positive correlation between the Attention Window Task and Game Creativity ($r = .25, p = .035$). Bayesian Analysis showed moderate support for the Alternative Hypothesis that the ability to spread attention widely is correlated with game creativity. See Fig. 3 for a visual illustration of the low associations between the cognitive ability measures and the Game Creativity Score.

5.3. Association between playing level and cognitive ability measures

Finally, we explored the relationship between the playing level of our sample of soccer players with the cognitive ability measures in Table 3 and Fig. 4. These analyses showed no significant correlations between the level the soccer players were playing at and how high they scored on the cognitive ability measures. Bayesian analyses indicated strong to moderate evidence in favor of H_0 : No relationship between playing level and cognitive ability measures.

6. Discussion

The main objective of the present research was to address shortcomings of previous studies investigating the frequently hypothesized association between domain-general cognitive abilities and domain-specific sport performance. To this end, we avoided well-documented pitfalls (Furley et al., 2023; Musculus et al., 2025) of typical cross-sectional between-group designs (Bonetti et al., 2025a for a recent high-profile example) by examining the correlation between carefully selected continuous measures of domain-general cognitive abilities (attentional breadth, multiple object tracking, working memory capacity, and inhibition) and continuous measures of domain-specific creative and intelligent soccer decision-making. Although it seems reasonable to expect that these domain-general cognitive abilities would support soccer decision-making, the overall pattern of results provides very limited support for this assumption as all correlation coefficients were small by conventional standards and Bayes analyses further indicated more support for the null hypothesis of no correlation. The only correlations that were significant after correcting for multiple comparisons was between multiple object tracking and game intelligence ($r = .38$) and attentional breadth and game creativity ($r = .25$). This suggests that soccer players with a better ability to track multiple objects simultaneously made slightly more intelligent decisions and soccer players with a greater attentional breadth made slightly more creative decisions. There was no association between playing level and any of the investigated cognitive ability measures as might have been assumed based on previous publications.

Taken together, the results do not provide strong support for the hypothesis that domain-general cognitive abilities—such as executive functions—substantially support domain-specific decision-making in soccer. Only two out of eight hypothesis tests supported this assumption: a positive association between multiple object tracking and game intelligence and between attentional breadth and creative decision-making. In this regard, the present study contributes to the critical sports cognition literature (Beavan et al., 2020a, 2020b; Fitzgerald et al., 2026; Furley et al., 2016, 2023; Furley & Memmert, 2010; Heisler et al., 2023; Knöbel et al., 2025; Memmert et al., 2009; Seidel-Marzi et al., 2026; Vona et al., 2024) and suggests that the influence of domain-general cognitive ability measures on domain-specific sport (soccer) performance is likely very limited. This finding aligns with key theoretical tenets from the expertise literature (e.g., Eccles, 2006; Ericsson et al., 2006) that people with years of experience in an activity such as soccer, only differ in cognitive skills directly related to their field of experience and that those skills do not translate to different domains. In this respect the present findings challenge the *cognitive component skill approach* (Voss et al., 2010) as domain-general ability measures that are associated with executive functions (i.e., working memory and inhibition ability) were not correlated with soccer game intelligence and creativity. In the present data, only multiple object tracking (MOT) and attentional breadth correlated with intelligent and creative passing decisions in soccer. Therefore, the ability to effectively spread visual attention and track multiple objects on a computer screen in a laboratory

Table 2
Correlations between cognitive ability measures and SoccerBot outcomes.

Cognitive Ability	SoccerBot_Task	<i>r</i>	<i>p</i>	<i>CI</i> _{Lower}	<i>CI</i> _{Upper}	<i>p</i> _{adj.}	<i>BF</i> ₀₁
Attention Window	Game Intelligence	.103	.263	−.078	.277	.789	4.739
Attention Window	Game Creativity	.252	.005	.077	.412	.035	.190
Working Memory	Game Intelligence	.147	.108	−.033	.318	.474	2.475
Working Memory	Game Creativity	.040	.637	−.127	.205	.999	8.547
Multiple Object Tracking	Game Intelligence	.378	.001	.171	.553	.008	.022
Multiple Object Tracking	Game Creativity	.154	.080	−.018	.318	.474	2.028
Inhibition low load	Game Intelligence	−.161	.079	−.331	.190	.474	1.905
Inhibition low load	Game Creativity	.028	.779	−.169	.224	.999	7.692

Note. *r* = Pearson correlation coefficient; *p* = unadjusted *p*-value; *CI*_{Lower} and *CI*_{Upper} = lower and upper bounds of the 95% confidence interval; *p*_{adj.} = Holm-Bonferroni adjusted *p*-value; *BF*₀₁ = Bayes factor quantifying evidence for the null versus the alternative hypothesis.

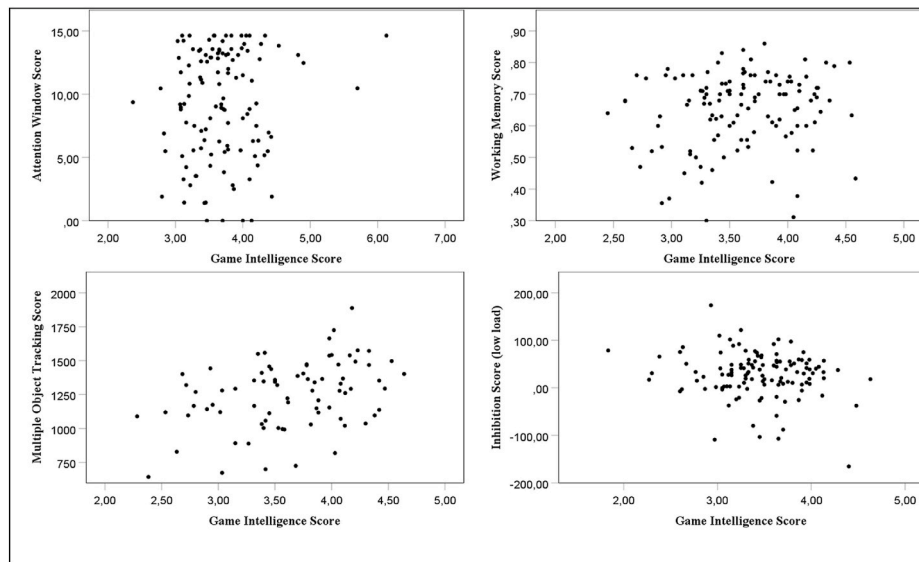


Fig. 2. Scatterplots illustrating the association between the main four cognitive ability measures and the SoccerBot game intelligence score.

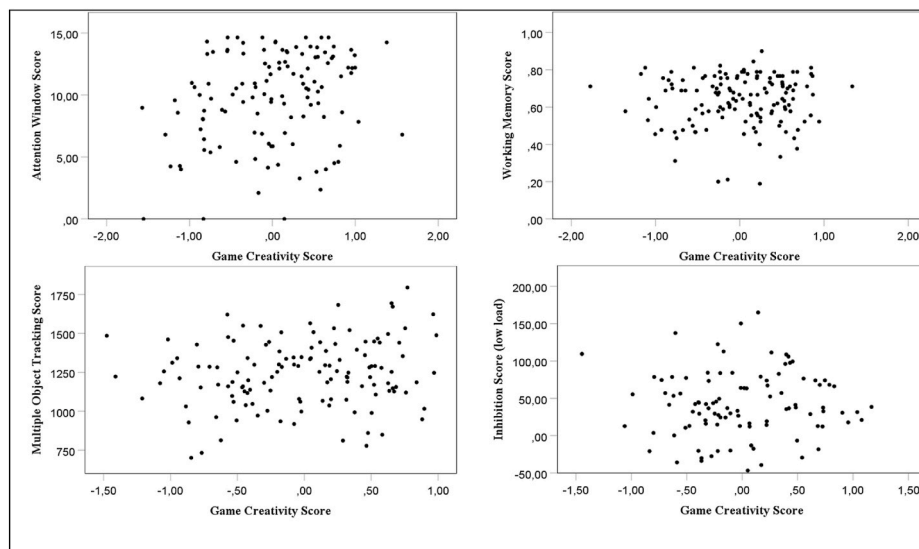


Fig. 3. Scatterplots illustrating the association between the main four cognitive ability measures and the SoccerBot game creativity score.

Table 3

Spearman Correlation of playing level with cognitive ability measures.

Cognitive Ability	n	Correlation	p_value	BF ₀₁
Attention window	194	.022	.765	10.526
Working Memory	241	.045	.486	6.135
Multiple Object Tracking	185	-.100	.176	1.802
Inhibition low load	201	.120	.090	4.329

Note. The n refers to the participant subgroup that completed the respective cognitive ability measure and for whom playing-level data was available.

setting was associated with better passing choices in the Soccerbot tasks. Although empirical evidence regarding the MOT-expertise link remains equivocal (Vater et al., 2021), according to the present study the MOT might show some promise as a laboratory measure to be associated with passing performance in soccer. Nevertheless, we acknowledge that more research is needed in this regard.

An additional point requiring consideration is the divergent pattern of findings for the MOT and perceptual load (inhibition) measures.

Whereas MOT performance showed a significant association with soccer-specific performance indices, the inhibition measure did not, although it was the only task that utilized stimuli that were (somewhat) soccer specific and therefore might arguably tap into soccer expertise. One plausible explanation is that MOT captures dynamic attentional allocation and the continuous tracking of multiple moving objects under time pressure, processes that resemble the perceptual-cognitive demands of soccer match play. In contrast, the inhibition task may primarily assess more static or domain-general attentional filtering mechanisms that are less directly aligned with the dynamic and interactive nature of in-game decision-making (despite the soccer-specific nature of the stimuli). Differences in task specificity may therefore explain why MOT exhibited associations with soccer-specific outcomes and the inhibition measure did not. At the same time, this pattern of results suggests that the relationship between perceptual-cognitive abilities and soccer performance is selective rather than general. Consequently, the present findings should not be interpreted as evidence that all attentional or perceptual-cognitive measures are equally associated with soccer-specific performance. Instead, the results might

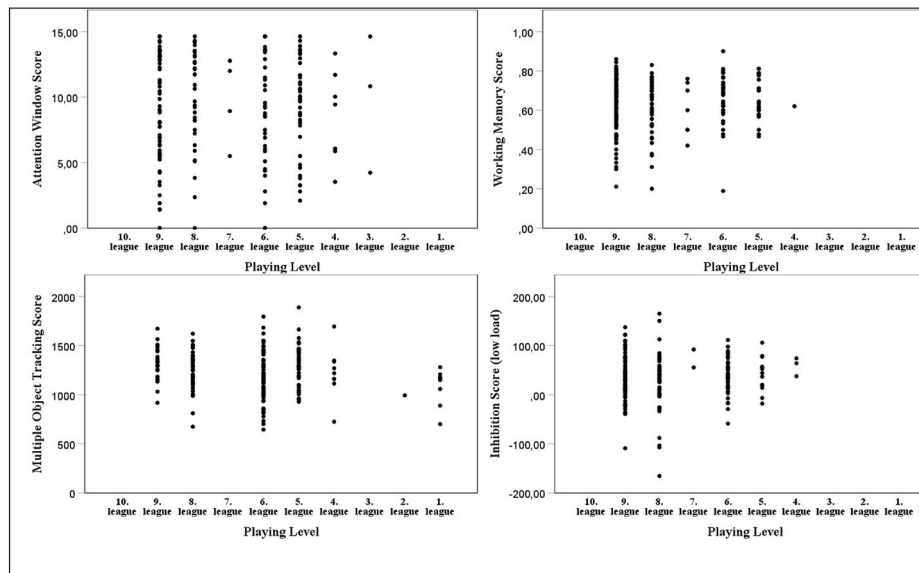


Fig. 4. Scatterplots illustrating the association between the main four cognitive ability measures and the playing level of the soccer players.

arguably support a more constrained interpretation, indicating that cognitive tasks sharing greater structural overlap with the temporal and attentional demands of soccer may show stronger performance associations.

The present findings also contribute to an ongoing, mixed literature regarding the role of domain-general cognitive abilities in team-sport expertise. While several meta-analyses have concluded that cognitive skills contribute to athletic expertise (Kalén et al., 2021; Logan et al., 2023; Scharfen and Memmert, 2021; Voss et al., 2010), other scholars have highlighted methodological challenges and publication biases that can influence primary studies in this field (Furley & Memmert, 2011; Simonsohn et al., 2022). Rather than serving as a direct corrective to meta-analytic conclusions, the modest and task-dependent associations observed here suggest that the relationship between domain-general cognition and sports performance may be more nuanced than previously assumed, underscoring the need for careful methodological scrutiny in future research.

Taken together, the present study in combination with previous work does have some applied implications regarding the use of domain-general cognitive ability tests in team-sports like soccer. For example, some authors have proposed that certain executive functions tests might have potential for talent identification in soccer (Vestberg et al., 2012) and the integration of executive function testing into talent identification protocols might offer a promising quantitative dimension to traditional scouting. Proponents argue that measures of executive functions provide an objective window into a player's anticipatory capacity and decision-making efficiency, which might be obscured by physical maturity in youth cohorts. However, together with a growing body of related research (Beavan et al., 2020a, 2020b; Furley et al., 2016, 2023; Furley & Memmert, 2010; Memmert et al., 2009; Vona et al., 2024), the present data does not support this practice. The predictive utility of domain-general cognitive assessments remains constrained by a lack of ecological validity; standardized, laboratory-based tasks may fail to capture the domain-specific representative design inherent in dynamic, open-skill sports like soccer. Consequently, if executive functioning profiles of soccer players can be utilized as a modular component of a holistic multidimensional battery for talent selection remains highly speculative and not sufficiently backed up by empirical data.

7. Strengths and limitations

The present research has several notable strengths and limitations.

As outlined throughout the manuscript, our main objective was to overcome key limitations in earlier studies on the link between domain-general cognitive abilities and domain-specific sport performance. A major strength is that we circumvented problems associated with between-group comparisons by employing a within-subject design using continuous variables for both cognitive abilities and soccer-specific performance. This approach minimized issues related to confounding variables, experimenter effects, and the loss of statistical power that often occurs when individuals are arbitrarily assigned to groups (albeit this problem was still evident when testing for correlations between playing level and domain-general cognition). Additionally, our study featured larger sample sizes and higher statistical power than most prior research in this area. We also improved upon earlier work by carefully selecting cognitive tests and correcting for multiple comparisons. That said, the study is not without limitations. First, our correlational analyses do not permit causal inferences. Second, we have to acknowledge that our large sample was scarce in the top league players, and the majority of soccer players were playing at a sub-elite level. Third, the study would have benefitted from having each participant complete all six tasks (four domain-general and two domain-specific) to further maximize power and allow for more robust within-subject comparisons. However, this was not feasible due to time constraints of the participants and one domain-general and one domain-specific test already took over 1 h of time. Arguably, the most substantial limitation stems from the soccer-bot decision making task. Although we used established measures of convergent (game intelligence; Memmert, 2010) and divergent thinking (creativity; Johnson & Raab, 2003; Memmert, 2011) in sports, the stimulus displays and extended decision times employed in the laboratory differ substantially from on-field soccer performance. This limitation applies particularly to the option-generation paradigm, which employed frozen, non-occluded game scenes paired with an extended response window (30 to 45 s). As soccer players do not have 45 s during match play to generate multiple options in a frozen game situation and subsequently rank-order them, the persistence of the visual display departs markedly from the dynamic, fleeting, and occluded nature of real-world sports environments. Nevertheless, this approach represents the primary established paradigm widely used in the option-generation and tactical creativity literature (cf. Hüttermann et al., 2019; Johnson & Raab, 2003; Memmert, 2010, 2011), and utilizing standard procedures reflects common scientific practice to maintain continuity and comparability within a given field of research. Rather than indexing rapid, automatic perception-action coupling or split-second tactical

recognition under severe time pressure, allowing the scene to remain visible likely recruits extended visual search, working-memory maintenance, and deliberate problem-solving. This methodological nuance has direct implications for how the resulting performance scores—and their associations with domain-general cognitive measures—are interpreted; specifically, cognitive tests that rely on rapid attentional disengagement or short-lived sensory representations may show weaker or more complex coupling with tasks that afford prolonged strategic contemplation. Moreover, our choice of cognitive ability measures might also be critiqued, although we paid attention to choose tasks with good psychometric properties and theoretical relevance to the field of soccer. We acknowledge that these design features may limit the generalizability of the present findings. Future research might therefore want to incorporate temporal occlusion paradigms or dynamic video simulations to better isolate fast perception-action processes from offline tactical reasoning.

8. Conclusion

In conclusion, the present research contributes to the growing body of work investigating the link between domain-general cognitive abilities and domain-specific performance in team sports. More specifically, we addressed a common limitation of prior research in this area, namely the reliance on cross-sectional between-group comparisons (e.g., experts vs. non-experts), which often confound cognitive performance differences with variations in training history, experience, or competitive level. Instead, we employed within-subject designs in comparatively large samples of at least 100 soccer players, allowing us to directly examine the relationship between domain-general cognitive abilities and soccer-specific decision-making skills within the same individuals. This approach reduces the influence of between-group confounds and provides a more sensitive test of whether interindividual differences in general cognitive functioning are meaningfully associated with soccer-specific perceptual-cognitive performance. While we observed a small-to-medium correlation between multiple object tracking and intelligent decision-making and a small correlation between attentional breadth and creative decision making in soccer, we found no significant associations in the remaining six hypothesis tests and no association between playing level and cognitive abilities. Taken together, these findings add to the growing skepticism about the role of domain-general cognitive abilities in explaining team-sport expertise.

CRedit authorship contribution statement

Philip Furley: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Justus Schlenger:** Writing – review & editing, Formal analysis, Data curation. **Sebastian Schwab:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Daniel Memmert:** Writing – review & editing, Supervision, Resources, Methodology, Data curation, Conceptualization.

Research disclosure statements

The authors declare that 1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in this manuscript; 2) that all independent variables or manipulations, whether successful or failed, have been reported in the manuscript; 3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in the manuscript/online supplements; 4) that (a) how sample size was determined and (b) our data-collection stopping rule have been reported in this manuscript.

Open science

All data will be made available at reasonable request to the corresponding author.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation and revision of this manuscript, we used Gemini (Google) to assist in refining and structuring the prose for clarity. Following the use of this tool, we reviewed, verified, and edited the content as needed and take full responsibility for the final content of the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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